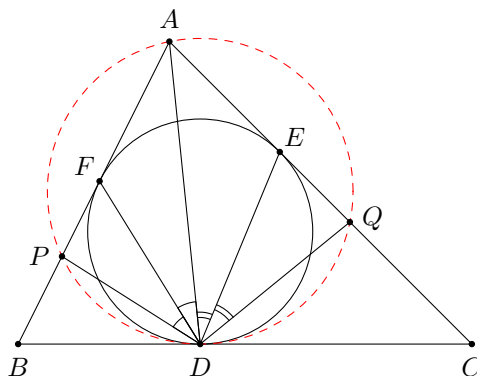


Problem 69 (Indonesia MO 2011 Problem 8). In $\triangle ABC$, its incircle touches the sides BC, CA, AB at D, E, F respectively. Let P be a point on the segment BF such that DF bisects $\angle ADP$. Let Q be a point on the segment CE such that DE bisects $\angle ADQ$. Prove that the circumcircle of $\triangle APQ$ is tangent to the incircle of $\triangle ABC$.



Solution. Since $\angle PDQ = 2\angle FDE = 2(180^\circ - \angle FDB - \angle EDC) = 360^\circ - 2\left(90^\circ - \frac{B}{2}\right) - 2\left(90^\circ - \frac{C}{2}\right) = B + C = 180^\circ - A$, the points A, P, D, Q are concyclic. Next, since

$$\angle PDB = \angle FDB - \angle FDP = \angle BFD - \angle ADF = \angle FAD = \angle PAD,$$

we know that BC is tangent to $(APDQ)$. Also, BC is tangent to (DEF) . So $(APDQ)$ and (DEF) are tangent at D . $\square$

Problem 70 (Dutch TST for BxMO/EGMO 2015 Problem 3). There is an $n \times n$ chessboard where $n \geq 2$. Find the largest positive integer k such that no matter how we colour the cells of the chessboard black or white, the following condition is always met: We can place k non-overlapping dominoes (of size 1×2 or 2×1) on the chessboard such that all of them cover two cells of the same colour, or all of them cover two cells of distinct colours.

Solution. Then answer is $\left\lfloor \frac{n^2}{4} \right\rfloor$.

Firstly, it is obvious that we can partition the chessboard into $\left\lfloor \frac{n^2}{2} \right\rfloor$ dominoes (with one extra cell when n is odd). By the pigeonhole principle, at least $\frac{1}{2} \left\lfloor \frac{n^2}{2} \right\rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor$ of these dominoes cover two cells of the same colour or cover two cells of distinct colours. So $k \geq \left\lfloor \frac{n^2}{4} \right\rfloor$.

Secondly, we colour a cell black if and only if it lies in an even row and an odd column. If we place k non-overlapping dominoes each covering cells of distinct colours, each domino must cover a black cell. So k cannot exceed the number of black cells, which is $\left\lfloor \frac{n^2}{4} \right\rfloor$. If we place k non-overlapping dominoes each covering cells of the same colour, the cells covered must be white since the black cells are not adjacent. Note that each such domino must cover a cell which lies in an odd row and an even column. There are $\left\lfloor \frac{n^2}{4} \right\rfloor$ such white cells. Therefore, we have $k \leq \left\lfloor \frac{n^2}{4} \right\rfloor$ in any case. The proof is complete. $\square$

Problem 71 (Rioplantense MO Level 3 2015 Problem 5). Find all positive integers d such that d is the greatest common divisor of

$$\binom{n+1}{n}, \binom{n+2}{n}, \dots, \binom{2n}{n}$$

for some positive integer n .

Solution. The answer is 1 or any prime number.

For any prime number p , if $n \neq p^s - 1$ for every positive integer s , then the base p representation of n is of the form $a_k \cdots a_1 a_0$ where not every digit is $p - 1$. Suppose $a_m < p - 1$ for some m . Then $1 \leq p^m \leq n$, and there is no carry when we compute the sum of n and p^m in base p . Therefore, $v_p \left(\binom{n+p^m}{n} \right) = 0$ by Kummer's theorem.

Therefore, if $n \neq p^s - 1$ for any prime number p , we must have $p \nmid d$ for all primes p , which means $d = 1$. Clearly, there exists such an n (e.g. $n = 5$).

It remains to consider $n = p^s - 1$ for some prime number p . In this case, for any $1 \leq j \leq n$, there must be a carry when we compute the sum of $n = ((p-1)(p-1) \cdots (p-1))_{(p)}$ and j in base p . This implies $v_p \left(\binom{n+j}{n} \right) \geq 1$ by Kummer's theorem. Also, when $j = p^{s-1}$, there is exactly one carry. So $v_p \left(\binom{n+p^{s-1}}{n} \right) = 1$. Therefore, we have $d = p$. $\square$

Problem 72 (Japan MO Finals 2019 Problem 3). Find all function(s) $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f \left(\frac{f(x)}{f(y)} + 1 \right) = f \left(\frac{x}{y} + y + 1 \right) - f(y) \quad (1)$$

for any $x, y \in \mathbb{R}^+$.

Solution. The answer is any function $f(x) = cx$ where c is a positive constant. We check that when $f(x) = cx$, then

$$f \left(\frac{f(x)}{f(y)} + 1 \right) = c \left(\frac{cx}{cy} + 1 \right) = c \left(\frac{x}{y} + y + 1 \right) - cy = f \left(\frac{x}{y} + y + 1 \right) - f(y).$$

Now, putting $x = y$ in (1), we have

$$f(2) = f(y+2) - f(y). \quad (2)$$

We prove that f is injective. Suppose on the contrary that $f(a) = f(b)$ for some $a > b$. Putting $x = a$ and $x = b$ in (1) respectively, we obtain

$$f \left(\frac{a}{y} + y + 1 \right) = f \left(\frac{f(a)}{f(y)} + 1 \right) + f(y) = f \left(\frac{f(b)}{f(y)} + 1 \right) + f(y) = f \left(\frac{b}{y} + y + 1 \right). \quad (3)$$

We choose y such that $\frac{a}{y} + y + 1 = \left(\frac{b}{y} + y + 1 \right) + 2$. This means we choose $y = \frac{a-b}{2} \in \mathbb{R}^+$. In that case, (3) yields a contradiction with (2) since $f(z+2) - f(z) = f(2) > 0$ for any z . Therefore, f must be injective.

Lastly, we put $y = 2$ in (1) and use (2) to obtain

$$f \left(\frac{f(x)}{f(2)} + 1 \right) = f \left(\frac{x}{2} + 3 \right) - f(2) = f \left(\frac{x}{2} + 1 \right).$$

By the injectivity, this implies $\frac{f(x)}{f(2)} + 1 = \frac{x}{2} + 1$, and hence $f(x) = cx$ for some constant $c = \frac{f(2)}{2}$. This completes the proof. $\square$